Maharashtra State Board Of Technical Education, Mumbai

Learning and Assessment Scheme Of Diploma Courses for Working Professionals

Programme Name	**: Diploma In Information Technology (Working Professional)**		
Programme Code	**: PIF**	**With Effect From Academic Year**	**: 2024-25**
Duration of Programme	**: 6 Semester**		
Semester	**: Fourth**	**Scheme**	**: F**

Sr No	Course Title	Abbrevation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme: Actual Contact Hrs./Week			Learning Scheme: Self Learning (Activity/ Assignment /Micro Project)	Learning Scheme: Notional Learning Hrs /Week	Credits	Assessment Scheme: Paper Duration (hrs.)	Theory				Based on LL & TL: Practical				Based on Self Learning		Total Marks
							CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
														Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
(All Compulsory)																								
1	DATABASE MANAGEMENT SYSTEM	DMS	DSC	313302	No	0	3	1	4	2	10	5	3	30	70	100	40	50	20	25#	10	25	10	200
2	DATA COMMUNICATION AND COMPUTER NETWORK	DCN	DSC	314318	No	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175
3	APPLIED MULTIMEDIA TECHNIQUES	AMT	SEC	313003	No	0	1	-	2	1	4	2	-	-	-	-	-	25	10	-	-	25	10	50
4	INTERNET OF THINGS	IOT	SEC	314006	No	0	1	-	4	1	6	3	-	-	-	-	-	25	10	25@	10	25	10	75
Total						**0**	**8**	**1**	**14**	**5**		**14**		**60**	**140**	**200**		**125**		**75**		**100**		**500**

Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :
1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)

Programme Name/s	**: Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Cloud Computing and Big Data/ Computer Technology/ Computer Engineering/ Computer Science & Engineering/ Data Sciences/ Computer Hardware & Maintenance/ Information Technology/ Computer Science & Information Technology/ Computer Science/ Electronics & Computer Engg./**
Programme Code	**: AI/ AN/ BD/ CM/ CO/ CW/ DS/ HA/ IF/ IH/ SE/ TE**
Semester	**: Third**
Course Title	**: DATABASE MANAGEMENT SYSTEM**
Course Code	**: 313302**

I. RATIONALE

This course focuses on fundamentals of relational database management system and enables students to design and manage database for various software applications. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in Information Technology applications.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

To design database and use any RDBMS package as a backend for developing database applications.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Explain concept of database management system.
- CO2 - Design the database for given problem.
- CO3 - Manage database using SQL.

- CO4 - Implement PL/SQL codes for given application.
- CO5 - Apply security and backup methods on database.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313302	DATABASE MANAGEMENT SYSTEM	DMS	DSC	3	1	4	2	10	5	3	30	70	100	40	50	20	25#	10	25	10	200

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

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2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain given database concept. TLO 1.2 Explain Overall structure of DBMS TLO 1.3 Describe architecture of database.	**Unit - I Introduction To Database System** 1.1 Database concepts:-Data, Database, Database management system, File system Vs DBMS, Applications of DBMS, Data Abstraction, Data Independence, Database Schema, The Codd's rules, Overall structure of DBMS 1.2 Architecture:- Two tier and Three tier architecture of database. 1.3 Data Models:- Hierarchical, Networking, Relational Data Models.	Presentations, Hands-on, Chalk-Board.
2	TLO 2.1 Explain relational structure of database. TLO 2.2 State types of keys with example. TLO 2.3 Draw ER diagrams for given problem. TLO 2.4 Explain different normalization forms.	**Unit - II Relational Data Model** 2.1 Relational Structure :- Tables (Relations), Rows (Tuples), Domains, Attributes, Entities 2.2 Keys :- Super Keys, Candidate Key, Primary Key, Foreign Key. 2.3 Data Constraints :- Domain Constraints ,Referential Integrity Constraints 2.4 Entity Relationship Model : - Strong Entity set, Weak Entity set, Types of Attributes, Symbols for ER diagram, ER Diagrams 2.5 Normalization:- Functional dependencies, Normal forms: 1NF, 2NF, 3NF	Presentations, Hands-on, Chalk-Board.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Write SQL queries using DDL, DML, DCL and TCL. TLO 3.2 Write SQL queries to join relations. TLO 3.3 Write SQL queries for ordering and grouping data. TLO 3.4 Use various class of operators in SQL. . TLO 3.5 Create schema objects for performance tunning.	**Unit - III Interactive SQL and Performance Tuning** 3.1 SQL: -Data-types, Data Definition Language (DDL), Data Manipulation language (DML), Data Control Language (DCL), Transaction Control Language (TCL). 3.2 Clauses & Join:- Different types of clauses - Where, Group by ,Order by, Having. Joins: Types of Joins, Nested queries. 3.3 Operators:- Relational, Arithmetic, Logical, Set operators. 3.4 Functions:- Numeric , Date and time, String functions, Aggregate Functions. 3.5 Views, Sequences, Indexes: -Views : Concept ,Create ,Update, Drop Views. Sequences :- Concept ,Create, Alter , Drop, Use of Sequence in table, Index: Concept ,Types of Index , Create ,Drop Indexes	Presentations, Hands-on, Chalk-Board.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Use control Structures in PL-SQL. TLO 4.2 Handle different types of exceptions. TLO 4.3 Explain various types of cursors. TLO 4.4 Create Procedure, Function on given problem. TLO 4.5 Explain types of triggers with examples	**Unit - IV PL/SQL Programming** 4.1 Introduction of PL/SQL: -Advantages of PL/SQL,The PL/SQL Block Structure, PL/SQL Data Types, Variable , Constant 4.2 Control Structure:- Conditional Control, Iterative Control, Sequential Control. 4.3 Exception handling: -Predefined Exception, User defined Exception. 4.4 Cursors:- Implicit and Explicit Cursors, Declaring, opening and closing cursor, fetching a record from cursor ,cursor for loops, parameterized cursors 4.5 Procedures:- Advantages, Create, Execute and Delete a Stored Procedure 4.6 Functions:- Advantages, Create, Execute and Delete a Function 4.7 Database Triggers :- Use of Database Triggers, Types of Triggers, Create Trigger, Delete Trigger	Presentations, Hands-on, Chalk-Board.

DATABASE MANAGEMENT SYSTEM **Course Code : 313302**

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Implement SQL queries for database administration. TLO 5.2 Explain concept of various types database backup processes. TLO 5.3 Describe various terms related to advanced database concepts.	**Unit - V Database Administration** 5.1 Introduction to database administration:- Types of database users, Create and delete users, Assign privileges to users 5.2 Transaction: Concept, Properties & States of Transaction 5.3 Database Backup: Types of Failures, Causes of Failure, Database backup introduction, types of database backups: Physical & Logical 5.4 Data Recovery – Recovery concepts , recovery techniques- roll forward ,Rollback 5.5 Overview of Advanced database concepts:- Data Warehouse ,Data lakes , Data mining, Big data ,Mongo DB , DynamoDB,	Presentations, Hands-on, Chalk-Board.

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Install database software	1	* Install the provided database software	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Create Database schema for given application	2	*Note :- Ensure to Carry out following activities before creating database: - Draw ER diagram for given problem - Normalize the relation up to 3NF 1) Create Database for given application 2) Create tables for the given application 3)Assign Primary key for created table 4) Modify the table as per the application needs	4	CO1
LLO 3.1 Execute DDL Commands to manage database using SQL	3	* Write queries using DDL Statements for following operations – 1)Create, alter, truncate, drop ,rename table 2) Apply Key Constraints for suitable relation.	2	CO3

DATABASE MANAGEMENT SYSTEM **Course Code : 313302**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Execute DML Commands to manipulate data using SQL	4	* Write queries using DML Statements for following operations – 1) Select, Insert, delete, update, table 2) Apply Key Constraints for suitable relation.	2	CO3
LLO 5.1 Execute DCL Commands to control the access to data using SQL .	5	* Write queries using DCL Statements for following operations – 1)Grant, Revoke	2	CO3
LLO 6.1 Execute TCL Commands to control transactions on data using SQL .	6	* Write queries using TCL Statements for following operations – 1) Commit, Rollback, Savepoint	2	CO3
LLO 7.1 Implement Queries using Arithmetic operators	7	Write Queries using built-in Arithmetic operators.	2	CO3
LLO 8.1 Implement Logical operators to apply various conditions in query.	8	Apply built-in Logical operators on given data	2	CO3
LLO 9.1 Implement Relational operators to apply various conditions in query.	9	Apply built-in relational operators on given data	2	CO3
LLO 10.1 Write Queries to implement SET operations using SQL .	10	* Use following Set operators to perform different operations.	2	CO3
LLO 11.1 Execute queries using String functions	11	Write SQL Queries using built-in String functions	2	CO3
LLO 12.1 Execute queries using Arithmetic functions	12	Write SQL Queries using built-in Arithmetic functions	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 13.1 Implement queries using Date and Time functions	13	Write Queries using built-in Date and Time functions	4	CO3
LLO 14.1 Implement queries using Aggregate functions	14	Write Queries using SQL built-in Aggregate functions	2	CO3
LLO 15.1 Execute Queries for ordering and grouping data.	15	* Implement Queries Using different Where, Having, Group by, & Order by clauses .	2	CO3
LLO 16.1 Execute the queries based on Inner & outer join	16	* Implement SQL queries for Inner and Outer Join	2	CO3
LLO 17.1 Create and manage Views for faster access on relations.	17	* Create and Execute Views ,Seqeuence and Index in SQL.	4	CO3
LLO 18.1 Implement PL/SQL program using Conditional Statements	18	* Write a PL/SQL program using Conditional Statements- if, if then else ,nested if, if elseif else	2	CO4
LLO 19.1 Implement PL/SQL program using Iterative Statements	19	* Write a PL/SQL program using Iterative Statements- loop,for, do-while, while	2	CO4
LLO 20.1 Implement PL/SQL program using Sequential Control	20	Write a PL/SQL program using Sequential Control-switch, continue,goto	2	CO4
LLO 21.1 Create implicit & explicit cursors	21	* Write a PL/SQL code to implement implicit & explicit cursors	2	CO4
LLO 22.1 Implement PL/SQL program based on Exception Handling (Pre-defined exceptions)	22	* Write a PL/SQL program based on Exception Handling (Pre-defined exceptions)	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 23.1 Implement PL/SQL program based on Exception Handling (user defined exceptions)	23	* Write a PL/SQL program based on Exception Handling (user defined exceptions)	2	CO4
LLO 24.1 Create Procedures and stored procedures for modularity.	24	* Write a PL/SQL code to create Procedures and stored procedures	2	CO4
LLO 25.1 Create function for given database	25	* Write a PL/SQL code to create functions.	2	CO4
LLO 26.1 Implement triggers for given database.	26	* Write a PL/SQL code to create triggers for given database.	2	CO4
LLO 27.1 Implement SQL queries for database administration.	27	Execute DCL commands using SQL 1) Create Users 2) Grant Privileges to users 3)Revoke Privileges to users	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Self Learning

- Implement PL/SQL code for relevant topics suggested by the teacher.

- Complete any one course related to Database Management System on Infosys Springboard platform.

Assignment

- Solve an assignment on any relevant topic given by the teacher.

Micro project

- Develop a database for restaurant management system. The restaurant maintain catalogue for the list of food items and generate bill for the ordered food.
- Prepare Invoice management system for electricity bill generation. Accept meter reading as inputs and generate respective bill amount for the same.
- Design a database for registration and admission of patient for Hospital management system, draw ER diagram and normalize the database up to 3NF.
- Any topic suggested by teacher.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Computer system - (Any computer system with basic configuration)	All
2	Any RDBMS software (MySQL/Oracle/SQL server/ or any other)	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction To Database System	CO1	6	4	6	2	12
2	II	Relational Data Model	CO2	8	2	4	6	12
3	III	Interactive SQL and Performance Tuning	CO3	12	2	6	10	18

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
4	IV	PL/SQL Programming	CO4	12	4	4	10	18
5	V	Database Administration	CO5	7	2	4	4	10
Grand Total				**45**	**14**	**24**	**32**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Continuous assessment based on process and product related performance indicators.
- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.
- A continuous assessment based term work.

Summative Assessment (Assessment of Learning)

- End semester examination, Lab performance, Viva voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	-	-	-	1	-	1			
CO2	2	2	3	2	1	2	1			
CO3	1	2	2	2	-	2	1			
CO4	1	3	3	2	1	3	2			
CO5	1	1	2	2	2	2	1			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Henry F. Korth	Database System Concepts	McGraw Hill Education ISBN : 9780078022159
2	Ivan Bayross	SQL, PL/SQL – The Programming Language of Oracle	BPB Publication ISBN 10: 8170298997 BPB Publication ISBN 13: 9788170298991

Sr.No	Author	Title	Publisher with ISBN Number
3	ISRD Group	Introduction to Database Management Systems	McGraw Hill Education ISBN 10: 0070591199 McGraw Hill Education ISBN-13 : 978-0070591196

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://nptel.ac.in/courses/106105175	Data Base Management System
2	https://www.w3schools.com/sql/	SQL Tutorial
3	https://www.tutorialspoint.com/sql/index.htm	SQL Programming Language

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

DATA COMMUNICATION AND COMPUTER NETWORK **Course Code : 314318**

Programme Name/s	**: Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Cloud Computing and Big Data/ Computer Technology/ Computer Engineering/ Computer Science & Engineering/ Data Sciences/ Computer Hardware & Maintenance/ Information Technology/ Computer Science & Information Technology/ Computer Science**
Programme Code	**: AI/ AN/ BD/ CM/ CO/ CW/ DS/ HA/ IF/ IH/ SE**
Semester	**: Fourth**
Course Title	**: DATA COMMUNICATION AND COMPUTER NETWORK**
Course Code	**: 314318**

I. RATIONALE

Data communication and computer networks are essential components of modern computing infrastructure, enabling seamless exchange of information and facilitating collaboration across various devices and locations. By considering various applications, students should be able to choose, classify, install, troubleshoot, and maintain various data communication networks. This course provides the important concepts and techniques related to networking and offer students to have valuable insights into technology behind network communication.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help the student to attain the following industry identified Outcome through various teaching learning experiences:

• Manage Data Communication and Computer Network

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Analyze the functioning of Data Communication and Computer Network.
- CO2 - Select relevant Transmission Media and Switching Techniques as per need.
- CO3 - Analyze the Transmission Errors with respect to IEEE standards.
- CO4 - Configure different TCP/IP services.
- CO5 - Implement relevant Network Topology using Networking Devices.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

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314318	DATA COMMUNICATION AND COMPUTER NETWORK	DCN	DSC	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs
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1	TLO 1.1 Describe the role of the given component in the process of data communication. TLO 1.2 Compare the characteristics of analog and digital signals on the given parameter. TLO 1.3 Explain the process of data communication using the given mode. TLO 1.4 Classify computer networks on the specified parameter.	**Unit - I Fundamentals of Data Communication and Computer Network** 1.1 Process of data communication and its components: Transmitter, Receiver, Medium, Message, Protocol 1.2 Protocols, Standards, Standard organizations, Bandwidth, Data Transmission Rate, Baud Rate and Bits per second 1.3 Modes of Communication (Simplex, Half duplex, Full Duplex) 1.4 Analog Signal and Digital Signal, Analog and Digital Transmission: Analog To Digital, Digital To Analog Conversion 1.5 Fundamental Of Computer Network: Definition And Need Of Computer Network, Applications, Network Benefits 1.6 Classification Of Network: LAN, WAN,MAN	Lecture Using Chalk-Board, Presentations, Video Demonstrations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain with sketches the construction of a given type of cable. TLO 2.2 Explain with sketches the characteristics of the given type of unguided transmission media. TLO 2.3 Explain with sketches the working of the given Multiplexing technique. TLO 2.4 Describe with sketches the working principle of the given Switching technique. TLO 2.5 Compare different Switching techniques on the given parameter.	**Unit - II Transmission Media And Switching** 2.1 Communication Media: Guided Transmission Media Twisted-Pair Cable, Coaxial Cable, Fiber-Optic Cable 2.2 Unguided Transmission Media: Radio Waves, Microwaves, Infrared, Satellite 2.3 Line-of-Sight Transmission, Point-to-Point, Broadcast 2.4 Multiplexing: Frequency-Division Multiplexing ,Time - Division Multiplexing 2.5 Switching: Circuit-switched network, Packet switched network	Lecture Using Chalk-Board, Presentations, Video Demonstrations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain working of the given error detection and correction method. TLO 3.2 Explain features of the given IEEE communication standard. TLO 3.3 Explain characteristics of the given layer in IEEE 802.11 architecture. TLO 3.4 Explain with sketches the process of creating a Bluetooth environment using the given architecture. TLO 3.5 Compare the specified generations of mobile telephone systems on the given parameter.	**Unit - III Error Detection and Correction** 3.1 Types of Errors, Forward Error Correction Versus Retransmission 3.2 Framing: Fixed Sized and Variable Sized Framing 3.3 Error Detection: Repetition codes, Parity bits, Checksums, CRC 3.4 Error Correction: Automatic Repeat Request (ARQ), Hamming Code 3.5 Wireless LAN IEEE 802.11 standard Architecture, Features of IEEE 802.11 versions: 802.11,802.11a,802.11b,802.11g,802.11n,802.11p 3.6 Bluetooth Architecture: Piconet, Scatternet 3.7 Mobile Generations: 3G, 4G and 5G	Lecture Using Chalk-Board, Presentations, Video Demonstrations, Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Identify functions and features of the given layer of OSI Reference model. TLO 4.2 Compare the specified service on the given parameters. TLO 4.3 Classify IP Addresses on the basis of its class from the given set of addresses. TLO 4.4 Distinguish between IPv4 and IPv6 on the given parameters. TLO 4.5 Describe with sketches the procedure to configure the given TCP/IP service.	**Unit - IV Network Communication Models** 4.1 THE OSI MODEL: Layered Architecture, Encapsulation 4.2 Layers in OSI Model(Functions of each layer)-Physical Layer,Data-Link Layer,Network Layer,Transport Layer,Session Layer,Presentation Layer,Application Layer 4.3 TCP/IP Layers and their functions: Host To Network Layer,Internet Layer,Transport Layer,Application Layer 4.4 Protocols: Host To Network Layer-SLIP,PPP, Internet Layer-IP,ARP,RARP,ICMP, Transport Layer-TCP and UDP, Application Layer-FTP,HTTP,SMTP,TELNET,BOOTP,DHCP 4.5 Addressing: Physical Address, Logical Address, Port Address 4.6 IP Address-Concept, Notation, Address Space 4.7 IPv4 Addressing: Classful and Classless Addressing ,subnet mask,supernetting,subnetting 4.8 IPV6 Addressing scheme and basic structure	Lecture Using Chalk-Board, Presentations, Case Study, Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Compare different computing models on the given parameter. TLO 5.2 Identify relevant network topology for the given situation. TLO 5.3 Compare different topologies on the given parameter. TLO 5.4 Select network connecting device for the given situation. TLO 5.5 Describe with sketches the procedure to configure the given networking device.	**Unit - V Network Topologies And Network Devices** 5.1 Network Computing Model: Peer To Peer, Client Server 5.2 Network Topologies: Introduction, Definition, Selection criteria, Types of Topology- Star ,Mesh, Tree, Hybrid 5.3 Network Connecting Devices: Switch, Router, Repeater, Bridge, Gateways and Modem	Lecture Using Chalk-Board, Video Demonstrations, Flipped Classroom

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Implement Amplitude Shift Keying(ASK)	1	* Amplitude Shift Keying(ASK) using any simulator	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Implement Frequency Shift Keying(FSK)	2	Frequency Shift Keying(FSK) using any simulator	2	CO1
LLO 3.1 Implement Phase Shift Keying(PSK)	3	Phase Shift Keying(PSK) using any open source simulation software	2	CO1
LLO 4.1 Create standard network straight cable by using cable tester.	4	*Create and Test standard straight network cable(Universal Colour Code) using crimping tool	2	CO2
LLO 5.1 Create standard Cross network cable by using cable tester.	5	Create and Test standard Cross network cable(Universal Colour Code) using crimping tool	2	CO2
LLO 6.1 Use basic programming skills to simulate communication systems. LLO 6.2 Debug and execute the program for Time Division Multiplexing(TDM).	6	* Generate a Time Division Multiplexing(TDM) signal using relevant simulation software	2	CO2
LLO 7.1 Transfer data using Bluetooth.	7	*Create a Hybrid Network Using Bluetooth	2	CO3
LLO 8.1 Identify different error detection methods. LLO 8.2 Detect errors using Checksum.	8	*Locate the error bit in the given data string by applying checksum error detection method	2	CO3
LLO 9.1 create WI-FI environment.	9	*Implement Wireless network	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Draw block diagram for parity check. LLO 10.2 Implement parity check with examples.	10	Write a 'C' program for parity check error detection	2	CO3
LLO 11.1 Implement C Program for CRC	11	*Write a 'C' program for Cyclic Redundancy Check(CRC) error detection	2	CO3
LLO 12.1 Implement Hamming code in any suitable programming language.	12	*Write a 'C' program for error correction using Hamming code	2	CO3
LLO 13.1 Use IP address and appropriate subnet mask for given problem statement.	13	*Configure static IP address in operating system along with appropriate subnet mask for given problem	2	CO4
LLO 14.1 Implement IP addresses for intranet in Class A, Class B, Class C.	14	* Implement Classful Address in a given network node i)Identify range of IP Address in various classes ii)Justify the reason to choose various IP address classes for creating given network	2	CO4
LLO 15.1 Troubleshoot computer network using commands.	15	*Execute TCP/IP network commands:ipconfig,ping,tracert	2	CO4
LLO 16.1 Troubleshoot computer network using commands.	16	*Execute TCP/IP network commands: netstat, pathping, route	2	CO4
LLO 17.1 Use wireshark packet sniffer software.	17	*1) Install Wireshark and configure as packet sniffer- i)Capture IP,TELNET, FTP packets using Wireshark	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Measure various types of Delay by using Wireshark.	18	Capture TCP and UDP packet using Wireshark	2	CO4
LLO 19.1 Filter ARP and ICMP packet Traffic using Wireshark.	19	Capture ARP and ICMP packet Traffic using Wireshark	2	CO4
LLO 20.1 Install server operating system	20	Install Operating System Linux/Windows/Any other Server	2	CO4
LLO 21.1 Create FTP Server	21	Use FTP protocol to transfer file from one system to another system	2	CO4
LLO 22.1 Implement IPv6 addressing scheme on a network.	22	Create IPv6 environment in a small network using simulator	2	CO4
LLO 23.1 Configure HTTP server on given operating system.	23	*Create HTTP server	2	CO5
LLO 24.1 Use star topology for a given situation.	24	*Create computers using Star topology with wired media	2	CO5
LLO 25.1 Use Network simulator CISCO packet tracer.	25	Create Tree topology using CISCO packet tracer software	2	CO5
LLO 26.1 Implement remote login feature.	26	Configure TELNET for remote login	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 27.1 Survey existing network infrastructure.	27	*Visit your computer laboratory- i)Identify the type of topology ii)Identify types of connecting devices with specifications iii)Identify types of cables with specifications iv)List the type of network applications commonly used in the laboratory iv)Draw the layout of installed network	4	CO5
LLO 28.1 Transfer a file from one computer to another. LLO 28.2 Print documents from remote system in a network.	28	Share folder and printer in a network	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- Solve an assignment on any relevant topic given by the Teacher
- For a trading firm an organization with 10users, draw network architecture design of wireless LAN.

- Identify appropriate network topology and network connecting devices for following requirement. Draw network design for proposed network. An organization having its office in a building of 5 floor. Each floor it needs 20 machines. There is one File server. Each floor has 2 print servers to facilitate printer capacity using Tree topology.

Micro project

- Install and configure NIC and find MAC Address of Device
- Design a network using any topology and do fault identification
- Create a tool that monitors network bandwidth usage in real-time

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Desktop Computer with basic configuration	All
2	Network Tool Kit: Crimping Tool for RJ-45 connector ,3in 1 modular crimping tool for RJ-45 UTP CAT-5/CAT-6 Networking Cable,LAN Cutter 8P/6pP/4P All-in-One or similar,Cable Tester/LAN Tester(Specification: Network Cable Tester for LAN RJ-45/CAT5/CAT6 UTP Wire Test Tool or similar)	All
3	Network Accessories: RJ45 connector, UTP cable, optical fibre cable, Coaxial cable, various connectors,1000Mbps NIC	All
4	UPS 6 KVA online	All
5	Ethernet Switch- 4/8/16/24/32	All
6	Router-256MB Memory storage capacity, compatible with Desktop and Laptop, Rack Mountable, Wireless Connectivity	All
7	Printer	All
8	Wireshark(https://www.wireshark.org/download.html)or any other Packet Analyzer Tool	All
9	Simulation Software: CISCO Packet Tracer, CORE Network Emulator or Similar	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Fundamentals of Data Communication and Computer Network	CO1	10	4	8	4	16
2	II	Transmission Media And Switching	CO2	10	4	4	6	14
3	III	Error Detection and Correction	CO3	8	4	4	6	14

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
4	IV	Network Communication Models	CO4	12	4	6	8	18
5	V	Network Topologies And Network Devices	CO5	5	2	2	4	8
Grand Total				**45**	**18**	**24**	**28**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Continuous assessment based on process and product related performance indicators.
- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.
- A continuous assessment based term work.

Summative Assessment (Assessment of Learning)

- End semester examination, Lab performance, Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	1	-	2	1	-	-	1			
CO2	1	1	2	1	-	1	1			
CO3	1	2	1	1	-	-	1			
CO4	1	2	2	1	-	1	1			
CO5	-	2	2	1	1	1	1			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Behrouz A. Forouzan	Data Communication and Networking	McGraw-Hill Higher Education ISBN-13 978-0-07-296775-3
2	Behrouz A. Forouzan:	TCP/IP Protocol Suit	McGraw Hill Education ISBN-13 978-0073376042

Sr.No	Author	Title	Publisher with ISBN Number
3	A.S. Tanenbaum	Computer Networks	PRENTICE HALL ISBN-10: 0-13-212695-8 ,ISBN-13:978-0-13-212695-3
4	Godbole Achyut	Data Communication and Networks	McGraw Hill Education ISBN-10 9780071077705,ISBN-13 978-0071077705
5	Comer Douglas E.	TCP/IP Principles, Protocols and Architectures	PEARSON ISBN 10: 0-13-608530-X ISBN 13: 978-0-13-608530-0

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.geeksforgeeks.org/data-communication-definition-components-types-channels/	Data Communication-Definition, Components,Types,Channels
2	https://www.tutorialspoint.com/data_communication_computer_network/index.htm	Data Communication and Computer Network
3	https://nptel.ac.in/courses/106105081	Computer Networks
4	https://nptel.ac.in/courses/106105183	Computer Networks and Internet Protocol
5	Introduction To Computer Networks | Studytonight	Introduction To Computer Networks

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Information Technology/ Computer Science & Information Technology**
Programme Code	**: IF/ IH**
Semester	**: Third**
Course Title	**: APPLIED MULTIMEDIA TECHNIQUES**
Course Code	**: 313003**

I. RATIONALE

Multimedia and Animation Techniques make connections between verbal and visual representations of contents. This practical oriented course help students to produce different components of Multimedia including text, images, audio, video and animation in order to use them in applications.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Construct different types of Multimedia.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Manipulate color models of image.
- CO2 - Perform edit operation on text and images using graphics processing tools.
- CO3 - Perform basic audio editing operations.
- CO4 - Perform basic video editing operations.
- CO5 - Create simple 2D Animation.
- CO6 - Design Web Pages with Multimedia components.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

APPLIED MULTIMEDIA TECHNIQUES Course Code : 313003

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313003	APPLIED MULTIMEDIA TECHNIQUES	AMT	SEC	1	-	2	1	4	2	-	-	-	-	-	25	10	-	-	25	10	50

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Define Multimedia and its applications. TLO 1.2 Describe types of display. TLO 1.3 Describe component of Multimedia. TLO 1.4 Describe various color models.	**Unit - I Introduction to Multimedia** 1.1 Definition of Multimedia, application of Multimedia: business, education. 1.2 Multimedia System Framework, Display System (LCD, LED, OLED, QLED, Foldable). 1.3 Component of Multimedia: text, graphics, audio, video and animation 2D and 3D. 1.4 Color models like RGB, CMYK, HSV, YIQ, saturation and brightness.	Demonstration Presentations Hands-on Lecture Using Chalk-Board
2	TLO 2.1 Convert text one form to another format. TLO 2.2 Describe different effects on text. TLO 2.3 Describe various image file formats. TLO 2.4 Compare Lossy and Lossless image compression techniques. TLO 2.5 Compare characteristics of 2D and 3D images.	**Unit - II Text and Image Editing** 2.1 Types of text format: plain text, RTF, PDF format. 2.2 Conversion of text one form to another format. Text effects (Ketchup, rope, Fire). 2.3 Graphics format: -Vector graphics formats: SVG, WMF, EPS, PDF, CDR -Raster Format: JPEG, PNG, TIFF, PNG, GIF, WebP, BMP and MPEG4. 2.4 Image compression techniques Lossy and Lossless. 2.5 Image effects: broken mirror effect, flaming ball effects, water drop effect. 2.6 2D and 3D images.	Demonstration Presentations Hands-on Lecture Using Chalk-Board

APPLIED MULTIMEDIA TECHNIQUES **Course Code : 313003**

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Describe features of given audio file formats. TLO 3.2 Compare Lossless vs Lossy compression.	**Unit - III Working with Audio** 3.1 Digital audio, Features of audio file formats: mp3, wav, mpeg7, mpeg21. 3.2 Lossless compressed audio format, Lossy compressed audio format. 3.3 MIDI, Mono, Stereo. 3.4 File Size.	Demonstration Presentations Flipped Classroom Hands-on
4	TLO 4.1 Explain digital video standards. TLO 4.2 Describe features of given video file format. TLO 4.3 Explain working of video streaming. TLO 4.4 Describe different types of Animations.	**Unit - IV Working with Videos and Animations** 4.1 Digital video, Broadcast video standards. 4.2 Video file formats: MPEG7, AVI, MP4, WMV. 4.3 Video Streaming: Introduction, Difference between streaming and downloading, working of streaming, buffering, factors affecting streaming. 4.4 Types of Animation: Object (Rolling Ball and Bouncing Ball) and Process animation, 2D, 3D, motion capture, motion graphics, morphing.	Presentations Video Demonstrations Hands-on Lecture Using Chalk-Board
5	TLO 5.1 Describe concept of action script. TLO 5.2 Write steps to develop a webpage and upload or publish web page. TLO 5.3 Explain concept of Virtual, Augmented and Mixed Reality.	**Unit - V Webpage Designing with Multimedia Components** 5.1 Programming concepts with respect to action script: variables, data types, conditionals, loops, arrays, functions. 5.2 Design web pages using Hypertext and Hypermedia. 5.3 Upload or publish web page. 5.4 Fundamentals and gadgets of Virtual, Augmented and Mixed Reality.	Presentations Video Demonstrations Presentations Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Implement different color models on image. LLO 1.2 Convert given image format into other format for optimum solution.	1	a. *Manipulate color related attributes of given images using any graphical processing tools on RGB, CMYK, HSV, YIQ color models. b. *Convert given image into different image formats, observe and report the changes in image with respect to quality and file size.	2	CO1 CO2
LLO 2.1 Implement various effects on text.	2	*Apply different effects on text using 2D image processing software such as: • Drop shadow • Mirror • Reflection	2	CO2
LLO 3.1 Implement different effects on image.	3	Apply different effects on GIF image using 2D image processing software such as: • Image mirroring • Rainy season effect	2	CO2
LLO 4.1 Create advertising banner.	4	Design advertising banner using graphics processing tools.	2	CO2

APPLIED MULTIMEDIA TECHNIQUES **Course Code : 313003**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 5.1 Create wallpaper showing water drop effect on image.	5	Design wallpaper showing water drop effect on GIF image using any 2D image processing software.	2	CO2
LLO 6.1 Design poster by using different text effect.	6	*Apply different effects on text to design poster using 2D image processing software such as: • Ketchup • Rope • Fire • Fruit	2	CO2
LLO 7.1 Implement given style on image.	7	Apply different style effects in JPEG image using 2D image processing software.	2	CO2
LLO 8.1 Implement Audio editing operations.	8	*Apply convert, merge, cut and join operation on digital audio files.	2	CO3
LLO 9.1 Implement Video editing operations.	9	Apply convert, merge, cut and join operation on video using video processing tool.	2	CO4
LLO 10.1 Apply shape twinning and motion effect in 2D animation.	10	Apply shape twinning and motion in 2D animation using 2D animation software.	2	CO5
LLO 11.1 Apply bouncing and rolling ball down effect in 2D animation.	11	*Apply bouncing and rolling ball down in 2D animation using 2D animation software.	2	CO5
LLO 12.1 Embed animation with audio into web page.	12	* Develop webpage which show animation with sound effect using any professional HTML5 editor.	2	CO3 CO6

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 13.1 Embed MP4 video into webpage.	13	* Develop webpage by embedding video using any professional HTML5 editor.	2	CO4 CO6
LLO 14.1 Embed Video Streaming on Web Page.	14	Develop a webpage for embedded video streaming using professional HTML5 editor.	2	CO4 CO6
LLO 15.1 Create simple animation.	15	*Create animation for rotating ball with action script using animation software such as Blender.	2	CO5 CO6
LLO 16.1 Apply Augmented Reality phenomena using relevant gadgets.	16	* Identify and experience Augmented Reality phenomena using gadgets such as smart phone/ google glass.	2	CO6

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- The microproject has to be industry application based, internet-based, workshop-based, laboratory-based or field-based as suggested by Teacher.
- Produce your college video for annual event.
- Design banner for departmental event.
- Develop interactive animated web page.

- Create animation clip for internal working of any one machine.
- Develop 2D animation clip for any cartoon story of 5 min.
- Produce 2D animation clip for advertising any product.

Other

- Explore information about mixed reality tools.
- Complete any one course related to multimedia on Infosys Springboard.
- Use ChatGPT/H20.ai or any other AI tool to explore information about image types and their differences.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Pencil 2D, OpenOffice draw, Microsoft Paint or any other such software.	1,5

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
2	Pencil 2D, Blender or any such software.	10,11
3	Pencil 2D, Blender or any such software Notepad++ or any advance HTML5 editor	12,13,14,15
4	Smart Phone / Google Glass or any other such device.	16
5	GIF Animator online tools: www.fotor.com any other such software.	2,3,4,7
6	Online Tool: https://flamingtext.com/ or any other such tool.	6
7	MP3 Cutter or any such software.	8
8	OpenShot Video Editor, Video Maker or any other such software.	9
9	Hardware: Computer (i3 onwards), with minimum 4GB RAM. Operating System: Windows 7/10/11 or Linux latest version.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction to Multimedia	CO1	2	0	0	0	0
2	II	Text and Image Editing	CO2	4	0	0	0	0
3	III	Working with Audio	CO3	2	0	0	0	0
4	IV	Working with Videos and Animations	CO4,CO5	4	0	0	0	0
5	V	Webpage Designing with Multimedia Components	CO6	3	0	0	0	0
Grand Total				**15**	**0**	**0**	**0**	**0**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Continuous assessment based on process and product related performance indicators.
- Each practical will be assessed considering 60% weightage to process 40% weightage to product.

Summative Assessment (Assessment of Learning)

- NA

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	1	-	-	1	-	-	-			
CO2	1	-	-	1	-	-	-			
CO3	1	-	-	1	1	-	1			
CO4	1	1	-	1	1	-	2			
CO5	1	2	1	2	1	1	2			

CO6	2	1	2	1	1	1	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Tay Vaughan	Multimedia: Making it work,9e	McGraw Hill Education, New Delhi 2015, ISBN:9780071832885
2	Parekh Ranjan	Principles of Multimedia 2e	McGraw Hill Education, New Delhi.2015, ISBN-13: 978-1-25-900650-0 ISBN-13: 1-25-900650-6
3	Colin Moock	Essential ActionScript 3.0	O'Reilly Media, Inc. ISBN: 9780596526948

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://helpx.adobe.com/in/animate/how-to/create-2d-animation.html	2D animation
2	https://photography.tutsplus.com/tutorials/learn-2d-animation-basics-in-blender--cms-41862	2D animation
3	https://www.gimp.org/tutorials/	Image editing
4	https://www.tutorialspoint.com/multimedia/	Multimedia concept
5	http://edutechwiki.unige.ch/en/AS3_Tutorials_Beginner	ActionScript
6	https://www.cloudflare.com/learning/performance/what-is-streaming/	Video Streaming

Sr.No	Link / Portal	Description
7	https://www.youtube.com/watch?v=vz0UUVDt2ps	Virtual Reality and Augmented Reality
8	https://drive.uqu.edu.sa/_/mskhayat/files/MySubjects/20178FS %20Multimedia%20Systems/Fundamentals_of_multimedia_2e.pdf	Fundamentals of Multimedia by Li, Ze-Nian

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s : **Information Technology/ Computer Science & Information Technology**

Programme Code : **IF/ IH**

Semester : **Fourth**

Course Title : **INTERNET OF THINGS**

Course Code : **314006**

I. RATIONALE

IoT is responsible for the super-fast evolution of industry 4.0, where the operations are mostly automated thus eliminating the need for much human intervention. The Internet of Things(IoT) describes the network of physical objects-“things” that are embedded with sensors , softwares and other technologies. IoT devices gather information and send it along to a data server where the information is collected, processed and used to make host of tasks easier to perform. IoT enables the creation of innovative solutions to real world challenges.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Develop and implement creative solutions for real time problems that can enhance efficiency, safety and convenience across various domains.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Integrate hardware and software for simple IoT applications.
- CO2 - Create IoT applications by interfacing various sensors and embedded boards.
- CO3 - Create IoT applications by interfacing various actuators and embedded boards.
- CO4 - Develop IoT applications using IoT networking devices.
- CO5 - Develop database based IoT application by integrating sensors with single board computer.

INTERNET OF THINGS Course Code : 314006

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
314006	INTERNET OF THINGS	IOT	SEC	1	-	4	1	6	3	-	-	-	-	-	25	10	25@	10	25	10	75

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe IoT building blocks with their relationship. TLO 1.2 State the features of various Embedded boards. TLO 1.3 Write simple Arduino program.	**Unit - I Basics of Internet of Things** 1.1 Introduction of IoT 1.2 Applications of IoT 1.3 Building blocks of IoT device: sensors, processors, gateways, applications 1.4 Embedded Boards for IoT: Arduino Uno, Raspberry pi, Node Microcontroller Unit, ESP32 1.5 Arduino Uno hardware architecture and Peripheral features 1.6 Arduino programming: Arduino programming of Blinking LED and Fading LED, Buzzer	Presentations Video Demonstrations Hands-on
2	TLO 2.1 Select various sensors for IoT applications. TLO 2.2 Write steps to interfacing sensors with Arduino TLO 2.3 Explain working techniques of Sensor.	**Unit - II Sensors in IoT** 2.1 Introduction of Sensors 2.2 IoT sensors types: Active and Passive Sensors, Analog sensors Digital Sensors 2.3 Programming with Arduino Sensors: Light sensor, Humidity Sensor, Temperature Sensor, Water Sensor, Motion Sensor, Fire Detection Sensors, Smoke Detection Sensors, Gas Detection Sensors, Soil moisture sensors 2.4 Basic working Technique of Sensor	Presentations Case Study Hands-on Demonstration Collaborative learning

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Select various Actuators available. TLO 3.2 Explain the process of interfacing appropriate actuator with Embedded boards. TLO 3.3 Write the steps for displaying output on various display devices.	**Unit - III Actuators in IoT** 3.1 Introduction of Actuators 3.2 Programming and Interfacing of actuators: displaying on LED/LCD with ATMEGA328 3.3 Displays: LCD, I2C LCD, 7 segment display 3.4 Actuators: Relay, Stepper Motor, Buzzer, Potentiometer	Presentations Demonstration Video Demonstrations Hands-on Model Demonstration
4	TLO 4.1 Explain IoT Protocols. TLO 4.2 Write the process to use IoT Wireless networking devices in developing IoT applications. TLO 4.3 Explain the method of performing Wi Fi connectivity to WEB.	**Unit - IV Communication in IoT devices** 4.1 Introduction to IoT networking: IoT Protocols- HTTP, MQTT, CoAP etc. 4.2 IoT Wireless devices and uses in IoT: LPWAN(Low Power Wide Area Networks),Cellular(3G/4G/5G), Bluetooth, Zigbee, Wi-fi, RFID 4.3 Wi Fi connectivity to WEB using ESP826	Presentations Demonstration Hands-on Video Demonstrations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Write the steps to install any operating system on Raspberry Pi. TLO 5.2 Write various Linux commands to be used on Raspberry Pi. TLO 5.3 Write steps to Install database on Raspberry Pi. TLO 5.4 Write database query to be performed on Raspberry Pi.	**Unit - V Programming with Single Board Computer** 5.1 Raspberry Pi Architecture, Features, Raspberry Pi Vs Arduino 5.2 Raspbian OS 5.3 Linux Programming Environment, Linux Commands 5.4 Installation of MariaDB server with Raspberry Pi	Presentations Demonstration Video Demonstrations Hands-on

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Install any embedded system. LLO 1.2 Write simple Arduino program using Arduino Uno IDE.	1	* Install any one embedded system(ex- Arduino IDE)and execute program to turn LED ON/OFF using delay	2	CO1
LLO 2.1 Interface RGB LED with Arduino. LLO 2.2 Write program to change the color of LED.	2	Change the color of LED	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Interface Potentiometer and LED with Arduino. LLO 3.2 Write a program to control the brightness of LED.	3	Control the brightness of LED using PWM Techniques	2	CO1
LLO 4.1 Interface LDR sensor with Arduino. LLO 4.2 Write a program for detection of Light.	4	* Detect the presence or absence of Light using LDR Sensor	2	CO2
LLO 5.1 Interface Analog Temperature Sensor (e.g. LM35) with Arduino. LLO 5.2 Write a program to sense Temperature of Object.	5	Measure the temperature of the object	2	CO2
LLO 6.1 Interface touch Sensor with Arduino. LLO 6.2 Write program to sense the touch when finger is placed on board.	6	Sense the touch of finger when it is placed on board	2	CO2
LLO 7.1 Interface IR Sensor with Arduino/ Raspberry Pi. LLO 7.2 Write program to Detect the obstacle.	7	Detect the obstacle using IR sensor	2	CO2
LLO 8.1 Interface Ultrasonic Sensor with Arduino/Raspberry Pi. LLO 8.2 Write a program to measure the Distance between sensor and object.	8	* Measure the Distance between sensor and object using ulatrasonic sensor	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Interface Gas Sensor with Arduino/ Raspberry Pi. LLO 9.2 Write program to detect the presence of Gas.	9	Detect the presence of Gas	2	CO2
LLO 10.1 Interface DHT11 sensor and I2C LCD with Arduino. LLO 10.2 Write a program to display Humidity and Temperature on LCD.	10	Detect the vibration of an object using vibration detector sensor SW-420 with Arduino	2	CO2
LLO 11.1 Interface PIR Sensor with Arduino/ Raspberry Pi to Detect Motion of object. LLO 11.2 Write a program to display motion detected or not.	11	Change the status of Buzzer ON/OFF	2	CO1 CO3
LLO 12.1 Interface DHT11 sensor and I2C LCD with Arduino. LLO 12.2 Write a program to display Humidity and Temperature on LCD.	12	* Display Humidity and Temperature on LCD using DHT11 sensor	2	CO2 CO3
LLO 13.1 Interface PIR Sensor with Arduino/ Raspberry Pi to Detect Motion of object. LLO 13.2 Write a program to display motion detected or not.	13	* Display the message as per detection of motion of object	2	CO2 CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 14.1 Interface DHT11 sensor and I2C LCD with Arduino. LLO 14.2 Write a program to display Humidity and Temperature on LCD.	14	* Control relay state based on input from IR sensor	2	CO2 CO3
LLO 15.1 Interface DHT11 sensor and I2C LCD with Arduino. LLO 15.2 Write a program to display Humidity and Temperature on LCD.	15	* Switch the LED ON/OFF on detection of obstacles using PIR sensor	2	CO2 CO3
LLO 16.1 Interface Ultrasonic Sensor, Buzzer with Arduino. LLO 16.2 Write program to start the buzzer when obstacle is detected in some specified range of distance.	16	* Measure the Distance between sensor and object and ring the buzzer when obstacle is detected in some specified range of distance	2	CO2 CO3
LLO 17.1 Interface Smoke Sensor with Arduino/ Raspberry Pi. LLO 17.2 Write program to detect smoke and play Burglar Alarm if smoke detected.	17	Play the Burglar Alarm if smoke detected	2	CO2 CO3
LLO 18.1 Interface Smoke Sensor with Arduino/ Raspberry Pi. LLO 18.2 Write program to detect smoke and play Burglar Alarm if smoke detected.	18	* Display percentage of moisture in soil using soil moisture sensor	2	CO2 CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 19.1 Interface fire detector sensor with NodeMCU. LLO 19.2 Write program to display to glow LED and play the alarm when fire detected.	19	Detect the fire and turn ON LED and play the alarm	2	CO2 CO3
LLO 20.1 Interface Ultrasonic Sensor with Arduino/Raspberry Pi. LLO 20.2 Write a program to measure the Distance between sensor and object.	20	* Display temperature value on serial monitor	2	CO3
LLO 21.1 Interface Piezo speaker with Arduino. LLO 21.2 Write program to play Melody.	21	* Play Melody sound with a Piezo speaker.	2	CO3
LLO 22.1 Interface Temperature sensor, Relay with Arduino. LLO 22.2 Write a program to turn it ON/OFF when Temperature increases or decreases.	22	* Control action using Relay based on temperature value	2	CO3
LLO 23.1 Interface seven segment display with Arduino. LLO 23.2 Write a program to display 0 to 9 numbers continuously .	23	* Display 0 to 9 numbers continuously on seven segment display	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 24.1 Interface I2C LCD with Arduino. LLO 24.2 Write program to display simple message.	24	Display simple message on I2C LCD	2	CO3
LLO 25.1 Interface Potentiometer and LCD with Arduino. LLO 25.2 Write program to display POT reading on LCD.	25	* Display POT value of potentiometer on LCD	2	CO3
LLO 26.1 Interface Bluetooth with Arduino/Raspberry Pi. LLO 26.2 Write a program to send sensor data to smartphone using Bluetooth.	26	* Transfer sensor collected data to smartphone using Bluetooth	2	CO4
LLO 27.1 Connect Camera module with Arduino. LLO 27.2 Write program to display the message on serial monitor when image is captured.	27	Display the message on serial monitor when image is captured	2	CO4
LLO 28.1 Connect temperature sensor with embedded board. LLO 28.2 Write program to display Temperature on Web Browser.	28	* Create Web based IoT application using Node MCU/Raspberry Pi to display Temperature on Web Browser	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 29.1 Install appropriate OS for embedded board. LLO 29.2 Connect various accessories to Rasberry Pi.	29	* Setup Raspberry Pi as an Single board computer b with following accessories: a display a cable to connect Raspberry Pi to display a keyboard a mouse SD card	2	CO5
LLO 30.1 Install database on single board computer. LLO 30.2 Perform various queries for displaying desired result.	30	* Install MariaDB database in Raspberry Pi and execute basic SQL queries	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Other

- Complete any one course related to "Internet of Things" freely available on Infosys Springboard/NPTEL/Spoken Tutorial.

Micro project

- Automatic Street Light- Street Light should automatically ON at evening and automatically OFF at morning. LCD and Serial Monitor shows Light Intensity value on First Line and Status of Street Light on Second Line. USE RGB LED for street Light and use orange color.

- Home Automation through PC- Design and develop project to control 8 home devices through PC serial monitor, LCD connected on project will shows Status of Devices is on or off. Also show the status of all devices on serial monitor.
- Motion enabled Room Light- Light present in Room should automatically ON when human motion is detected and automatically OFF in the absence of human motion. LCD and Serial monitor shows appropriate message as "Motion detected! Light ON" and "No Motion! Light OFF" when particular condition fulfilled.
- Electronic Smart Blind Stick- If someone is in front of blind person, LED and Buzzer should on and LCD will show the message" Obstacle. Be Alert" otherwise LED and Buzzer will remains off and LCD show the message "Safe.. Keep Walking".
- Electronic Notice Board- Any Message send from Serial Monitor should get displayed on LCD. When new message sends, previous message gets automatically erased and replaced with new message.

Assignment

- Solve Assignment covering all COs given by Course Teacher.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Sensors-LDR,IR,PIR,Ultrasonic Sensor,DHT11,LM35, touch sensor,smoke sensor, gas detect sensors(CO2,O2,NO2 etc.), fire sensor	20,6,12,13,14,7,15,8,16,9,17,10,11,18,19,21,22,23,24
2	Bluetooth, Wi-Fi,Ethernet modules	26,28
3	Arduino/NodeMCU/Raspberry Pi-controllers	All
4	Actuators- LED, Buzzer, Swiches, Relay,Sprinkler , I2C LCD, 7 segment display, potentiometer, Servo motor, Stepper motor, DC motor, Camera module	All
5	Accessories - Resistors, Jumper wires, Bread Board	All
6	Software tools-Arduino UNO IDE, Tinkercad, Linux	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Basics of Internet of Things	CO1	3	0	0	0	0
2	II	Sensors in IoT	CO2	4	0	0	0	0
3	III	Actuators in IoT	CO3	3	0	0	0	0
4	IV	Communication in IoT devices	CO4	3	0	0	0	0
5	V	Programming with Single Board Computer	CO5	2	0	0	0	0
Grand Total				**15**	**0**	**0**	**0**	**0**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Each Practical will be assessed considering 60% weightage to the process, 40% weightage to the product.

Summative Assessment (Assessment of Learning)

- End Semester Exam based on Practical performance and Viva-voce.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	1	2	2	1	2	1			
CO2	1	2	2	3	2	2	2			
CO3	1	2	2	3	2	2	2			

CO4	1	2	2	3	2	2	2			
CO5	2	2	3	3	2	2	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Cornel M Amariei	Arduino Development Cookbook	PACKT publishing Ltd. New Delhi, ISBN: 978-1-78398-294-3
2	Arshdeep Bahga, Vijay Madisetti	Internet of Things: A Hands-On Approach	Orient Blackswan New Delhi ,ISBN: 978- 0996025515 628/- 2
3	David Hanes, Gonzalo Salgueiro, Patrick Grossetti	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things	Cisco Press ISBN: 978-1- 58714-456-1 599
4	Simen Monk	Raspberry Pi Cookbook	Publisher(s): O'Reilly Media, Inc. ISBN: 9781098130923
5	Agus Kurniawan	Smart Internet of Things projects	PACKT publishing Ltd. New Delhi ISBN:9788131766613

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://github.com/microsoft/IoT-For-Beginners	All practicals
2	https://www.javatpoint.com/difference-between-sensors-and-actuators	Sensors and Actuators

Sr.No	Link / Portal	Description
3	https://www.tinkercad.com/learn/circuits?collectionId=O0K87SQL1W5N4P2	Practical using Simulator
4	https://www.geeksforgeeks.org/introduction-to-internet-of-things-iot-set-1/	Online content of Internet of Things
5	https://hands-on-books-series.com/iot.html	Introduction to IoT
6	https://www.raspberrypi.org/	Raspberry Pi Hands on tutorial

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students